Note- You should not include a separate Test variable because Test 1 and Test 2 are not independent variables. They are **replicates** of the same experiment. In a statistical context, each row of your data table should represent a single, independent observation. In your experiment, each measurement of the zone of inhibition is an individual observation.

The goal of the Anova Test is to is to determine if there is an **overall significant difference** in the mean zone of inhibition across all the different concentrations of citronella oil.

>**Null Hypothesis (H0)**: There is no significant difference in the mean zone of inhibition across the different citronella oil concentrations.

>**Alternative Hypothesis (Ha)**: There is a significant difference in the mean zone of inhibition across the different citronella oil concentrations.

One way Anova.

Descriptives

Zone of Inhibition

	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean		Minimum	Maximum
					Lower Bound	Upper Bound		
Positive Control	4	18.00	.816	.408	16.70	19.30	17	19
50%	4	13.75	1.500	.750	11.36	16.14	12	15
25%	4	9.00	.816	.408	7.70	10.30	8	10
12.5%	4	8.00	.816	.408	6.70	9.30	7	9
Total	16	12.19	4.230	1.058	9.93	14.44	7	19

As shown in the descriptive statistics table, the mean zone of inhibition for the 50% concentration was 13.75 mm, while the 25% concentration had a mean of 9.00 mm, and the 12.5% concentration had a mean of 8.00 mm, this suggests a direct relationship between citronella oil concentration and antibacterial activity, with higher concentrations producing larger zones of inhibition.

ANOVA

Zone of Inhibition

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	255.688	3	85.229	80.216	.000
Within Groups	12.750	12	1.063		
Total	268.438	15			

The P-Value of 0.000 is less than the p value of 0.05, this shows that there is a highly significant difference in the mean zone of inhibition across the different concentrations of citronella oil. A p-value of 0.000 provides strong evidence that the concentration of the oil is the reason for the different levels of antibacterial activity, not luck.

Multiple Comparisons

Dependent Variable: Zone of Inhibition

Tukey HSD

(I) concentration of the citronella oil.	(J) concentration of the citronella oil.	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval	
					Lower Bound	Upper Bound
Positive Control	50%	4.250*	.729	.000	2.09	6.41
	25%	9.000*	.729	.000	6.84	11.16
	12.5%	10.000*	.729	.000	7.84	12.16
50%	Positive Control	-4.250*	.729	.000	-6.41	-2.09
	25%	4.750*	.729	.000	2.59	6.91
	12.5%	5.750*	.729	.000	3.59	7.91
25%	Positive Control	-9.000*	.729	.000	-11.16	-6.84
	50%	-4.750*	.729	.000	-6.91	-2.59
	12.5%	1.000	.729	.539	-1.16	3.16
12.5%	Positive Control	-10.000*	.729	.000	-12.16	-7.84
	50%	-5.750*	.729	.000	-7.91	-3.59
	25%	-1.000	.729	.539	-3.16	1.16

*. The mean difference is significant at the 0.05 level.

This Post Hoc Tests table, specifically the Tukey HSD test is used to pinpoint exactly where the significant differences lie, since your ANOVA result was significant ($p = 0.000$). The Sig. value is used to determine which comparisons are significant.

1. **Positive Control vs. others**: All comparisons involving the Positive Control are highly significant ($p = 0.000$). This means the Positive Control has a significantly larger zone of inhibition than all other concentrations.

2. **50% vs. others**: The 50% concentration is significantly different from the Positive Control ($p = 0.000$), 25% ($p = 0.000$), and 12.5% ($p = 0.000$). This shows the 50% concentration is much more effective than the lower concentrations.

3. **25% vs. 12.5%**: The Sig. value for this comparison is 0.539. Since $0.539 > 0.05$, there is no statistically significant difference between the antibacterial activity of the 25% and 12.5% concentrations.

In conclusion, all concentrations of citronella oil are significantly less effective than the positive control, the 50% concentration is significantly more effective than both the 25% and 12.5% concentrations. There is no significant difference in effectiveness between the 25% and 12.5% concentrations.

Homogeneous Subsets

Zone of Inhibition

Tukey HSD

concentration of the citronella oil.	N	Subset for alpha = 0.05		
		1	2	3
12.5%	4	8.00		
25%	4	9.00		
50%	4		13.75	
Positive Control	4			18.00
Sig.		.539	1.000	1.000

Means for groups in homogeneous subsets are displayed.

a. Uses Harmonic Mean Sample Size = 4.000.

Based on the Homogeneous Subsets table, the results show that the different concentrations of citronella oil have varying levels of antibacterial activity, which can be grouped into three distinct categories.

1.The **12.5%** and **25%** concentrations are in the same group because their effects on the zone of inhibition are not significantly different from each other.

2.The **50%** concentration is in its own group, meaning it is significantly more effective at inhibiting bacterial growth than the 12.5% and 25% concentrations.

3.The **Positive Control** is also in its own group, demonstrating that it is significantly more effective than all of the citronella oil concentrations.